

Object Oriented Programming

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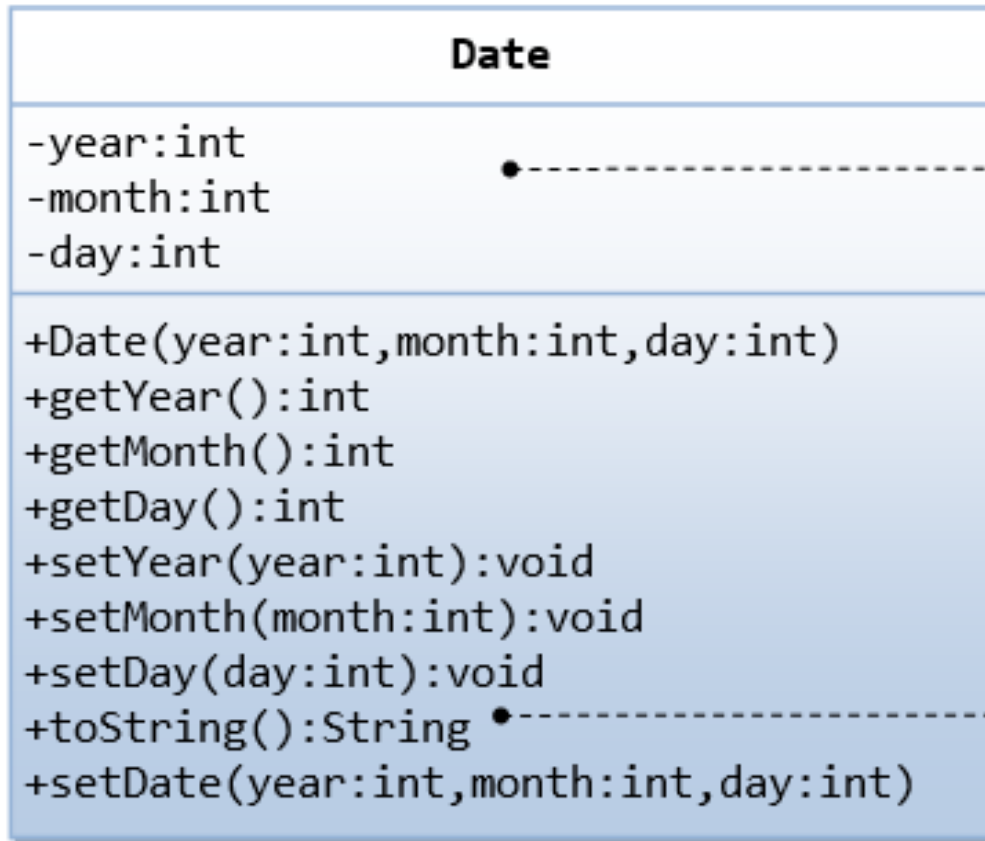
Last Week

- Object and classes
 - Default constructor
 - Copy constructor
 - Out of class function definitions
 - .h and .cpp file for a class
 - User defined String class
 - Buffer management
 - Extra functionality
 - STL **string** class

This Week

- Operator overloading

Class Diagram Review



day: [1, 31]
month: [1, 12]
year: [1000, 9999]

"MM/DD/YYYY" with leading zero

Quiz

Operator overloading

- What are operators?
- What is overloading?
- Operator overloading: Customizes the C++ operators for operands of user-defined types.

Operator overloading Examples

- Distance class:
 - `d3 = d1.addobjects(d2);`
 - `d3 = d1 + d2;`
- Complex: `c3 = c1 + c2;`
- Matrix: `m3 = m1 + m2;`

Counter class Example

- Overload unary operators ++, -- to increment, decrement counter
- How about `c2 = ++c1`?

Complete class example

- Complex numbers: $a + bi$
 - Real part
 - Imaginary part
- Draw the class diagram and create the Complex class. Write a constructor to construct a complex object. Write and add function to add two Complex objects. Write a print function to display the complex variable.

Operator overloading in Complex class

- Overload + operator to perform addition
 - // This is automatically called when '+' is used with
 - // between two Complex objects
 - Complex operator + (Complex const &obj) {
 - Complex res;
 - res.real = real + obj.real;
 - res.imag = imag + obj.imag;
 - return res;
 - }

Distance class revisited

- Apply operator overloading to distance class.
- How about comparing two distances?

Overloadable operators

+	-	*	/	%	^	&		~
!	=	<	>	+=	-=	*=	/=	%=
^=	&=	=	<<	>>	<<=	>>=	==	!=
<=	>=	&&		++	--	,	->*	->
()	[]	new	delete	new[]	delete[]			

Summary

- Operator overloading is an exciting C++ concept
- Improves program readability

Next Week

- Operator overloading Contd